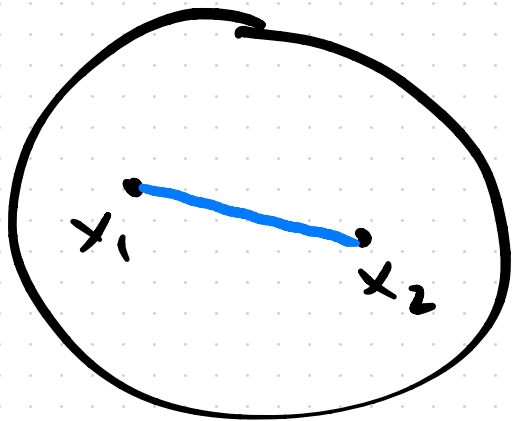
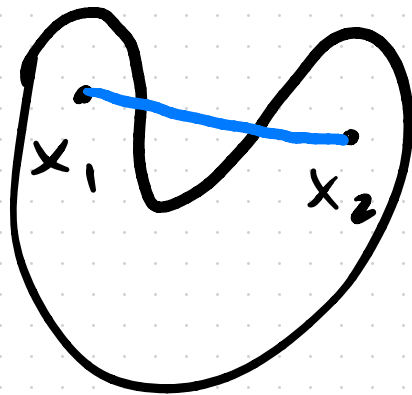


Convex sets

Let $V = \mathbb{R}^n$ be a finite dim. real vector space. A set $K \subset V$ is **convex** if for all $x_1, x_2 \in K$ and $0 < \lambda < 1$,
 $(1-\lambda)x_1 + \lambda x_2 \in K$.



convex



not convex

Example: Let $y_1, y_2, \dots, y_m \in V^*$, that is, $y_i: V \rightarrow \mathbb{R}$ is a linear functional. Let $b_1, \dots, b_m$ be real numbers. Then

$$\{x \in V: y_i(x) \leq b_i \text{ for all } i\}$$

is convex. Any set of this form is called a **polyhedron**.

In fact, if $\{y_i\}_{i \in I}$ is some infinite set of linear functionals and $b_i \in \mathbb{R}$ for all $i \in I$, then

$$\{x \in V : y_i(x) \leq b_i, i \in I\}$$

is convex. However, for a polyhedron we require that it can be represented with finitely many inequalities.

Given points $x_1, \dots, x_m \in V$, a **convex combination** is a sum of the form

$$\sum_{i=1}^m \lambda_i x_i, \quad 0 \leq \lambda_i \leq 1, \quad \sum_{i=1}^m \lambda_i = 1$$

The **convex hull** of a set $S \subset V$ is the smallest convex set containing S . Denoted **conv** S .

Equivalently, it is the set of all convex combinations of finite subsets of S .

Example: let $x_1, \dots, x_{n+1}$ be **affinely independent** points in $\mathbb{R}^n$, i.e. if

$$\sum_{i=1}^{n+1} \lambda_i x_i = 0$$

$$\sum_{i=1}^{n+1} \lambda_i = 0$$

then all $\lambda_i = 0$.

Then $\text{conv}(x_1, \dots, x_{n+1})$ is a **simplex** of dimension n .

$$n=0$$



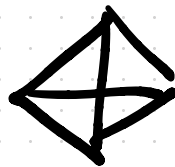
$$n=1$$



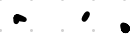
$$n=2$$



$$n=3$$



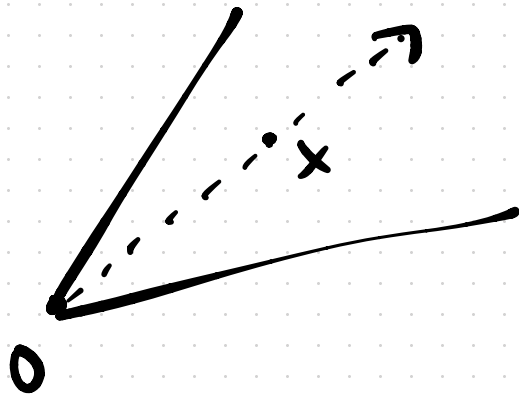
$$n=4$$



A **cone** is a set $C \subset V$ such that

(1) $0 \in C$

(2) If $x \in C$, then $\lambda x \in C$ for any $\lambda \geq 0$.



Given vectors $x_1, \dots, x_m \in V$, a **conical combination** is a sum of the form.

$$\sum_{i=1}^m \lambda_i x_i, \quad \lambda_i \geq 0$$

The **conical hull** of a set $S \subset V$ is the smallest convex cone containing S . Denoted **cone S** .

Equivalently, it is the set of all conical combinations of finite subsets of S .

Example: Let $x_1, \dots, x_n$ be linearly independent vectors in $\mathbb{R}^n$.

Then $\text{cone}(x_1, \dots, x_n)$ is called a **simplicial cone**.

$n=0$



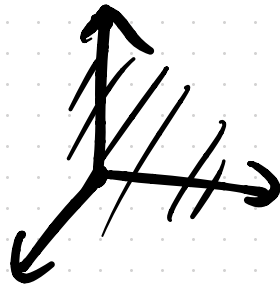
$n=1$



$n=2$



$n=3$



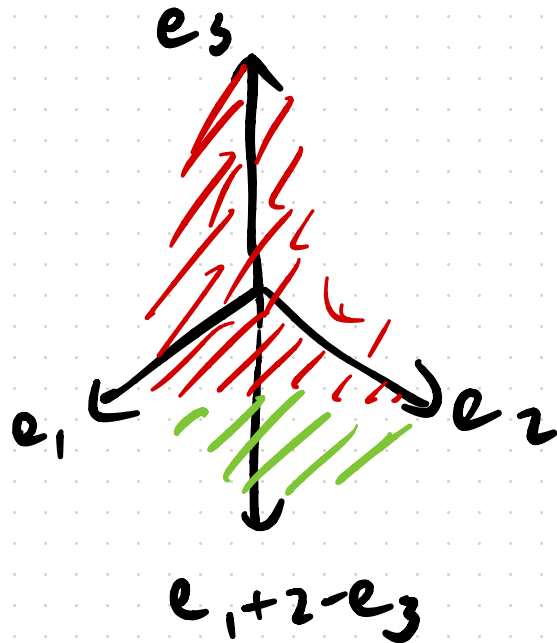
$n=4$

A simplicial cone is linearly equivalent to the nonnegative orthant in $\mathbb{R}^n$.

$$\mathcal{C} \\ \{x \in \mathbb{R}^n, x_i \geq 0\}$$

Example: Let standard basis vectors in $\mathbb{R}^3$

$$C = \text{cone}(e_1, e_2, e_3, e_1 + e_2 - e_3)$$



$$C = \text{cone}(e_1, e_2, e_3)$$

$$\cup \text{cone}(e_1, e_2, e_1 + e_2 - e_3)$$

Carathéodory's Theorem: Let $S \subset V = \mathbb{R}^n$. If $x \in \text{core } S$, then $x \in \text{core } T$ for some $T \subset S$ with $|T| \leq n$.

Proof: We can write $x = \sum_{i=1}^m \lambda_i x_i$ where $x_1, \dots, x_m \in S$, $\lambda_i > 0$. If $m \leq n$, we are done. Suppose $m > n$. Then $x_1, \dots, x_m$ are linearly dependent. Let

$$\sum_{i=1}^m \mu_i x_i = 0 \quad \mu_i \text{ not all } 0$$

We can assume some $\mu_i > 0$.

$$x = \sum_{i=1}^m \lambda_i x_i - \alpha \sum_{i=1}^m \mu_i x_i \quad \alpha > 0$$

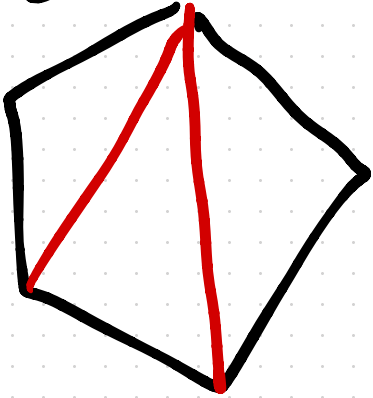
$$= \sum_{i=1}^m (\lambda_i - \alpha \mu_i) x_i$$

Start α at 0 and move to ∞ . At some point, one of the $\lambda_i - \alpha \mu_i$ will equal 0 and the rest will be ≥ 0 . Removing this x_i , we have that x is a conical combination of $m-1$

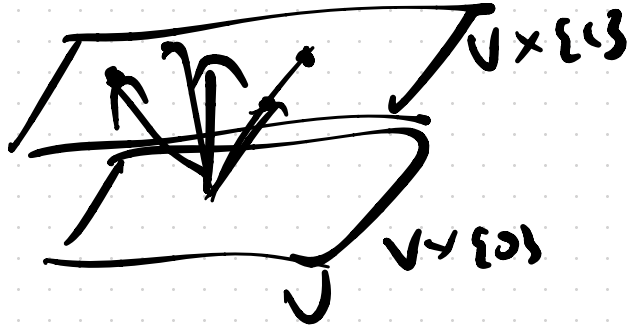
element of S . Repeat.

Corollary: Let $S \subset \mathbb{R}^n$. If $x \in \text{conv} S$, then $x \in \text{conv} T$ for some $T \subset S$ and $|T| \leq n+1$.

$n=2$



Proof: Apply previous theorem to the set $S \times \{1\}$ in $U \times \mathbb{R}$.



Corollary: The conical or convex hull of a finite set closed.

Proof: We induct on the dimension of V . Let S be a finite set in V . By previous theorem,

$$\text{cone } S = \bigcup_{\substack{T \subseteq S \\ |T| \leq \dim V}} \text{cone } T$$

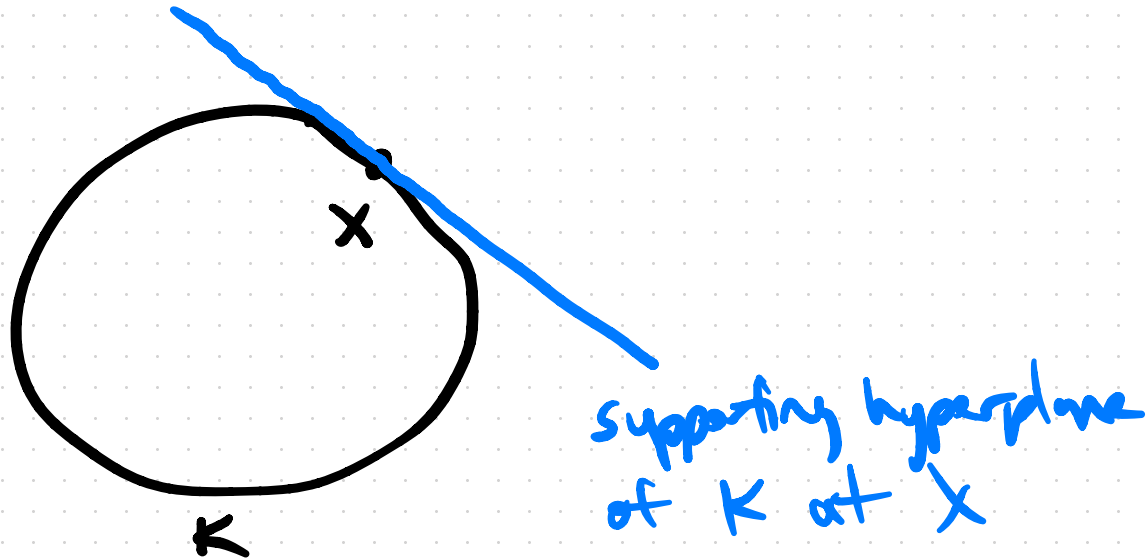
finite union

Suffices to prove each $\text{cone } T$ is closed.

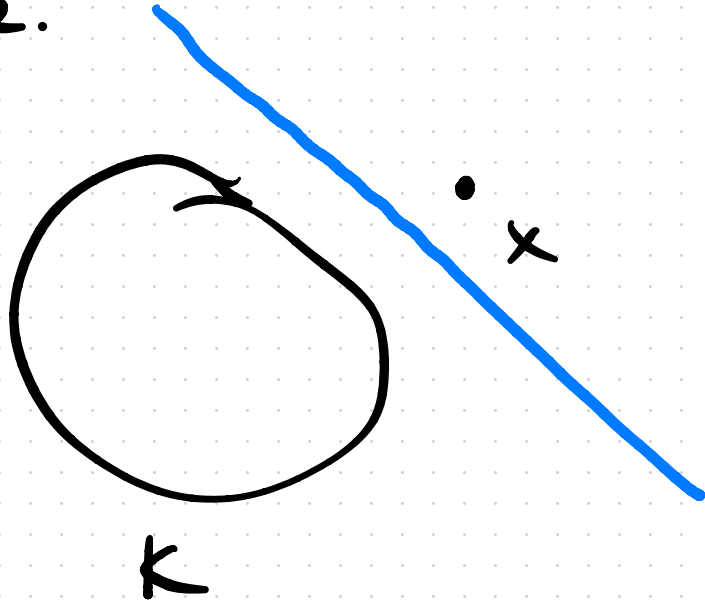
If $\text{span } T = V$. Then $|T| = \dim V$, and T is a linearly independent set of vectors. Thus $\text{cone } T$ is a simplicial cone, which is closed.

If $\text{span } T < V$. Then $T \subset W$ for some proper subspace W of V . Then $\text{cone } T$ is closed by induction.

Supporting hyperplane theorem: Let K be a closed convex set, and let x be on the boundary of K . Then there is a hyperplane passing through x so that K is weakly on one side of the hyperplane.



Separating hyperplane theorem: Let K be a closed convex set, and let $x \notin K$. Then there is a hyperplane such that x and K are strictly on opposite sides of the hyperplane.



Lemma: Let C be a convex cone which is not all of V . Then there is a hyperplane supporting C at 0 .

Proof:

